

# The Influence of Demographic Factors on Medical Insurance Charges: A Dive Into Sex and Age

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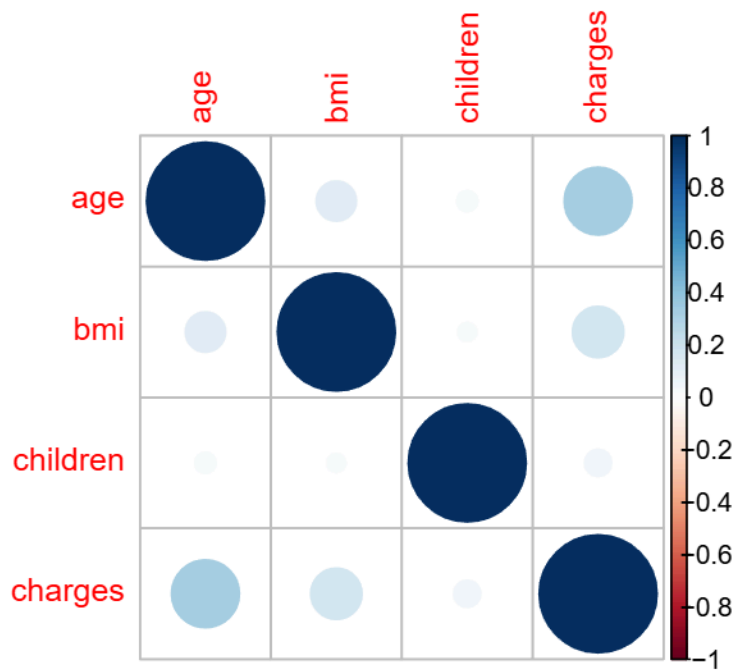
## Introduction

Medical insurance costs affect a lot of people, from individuals and families to healthcare providers themselves, impacting both access to care and personal finances. Figuring out what drives these costs allows companies to set fair insurance premiums and help people plan for healthcare expenses. This project looks into how age and sex influence medical insurance charges.

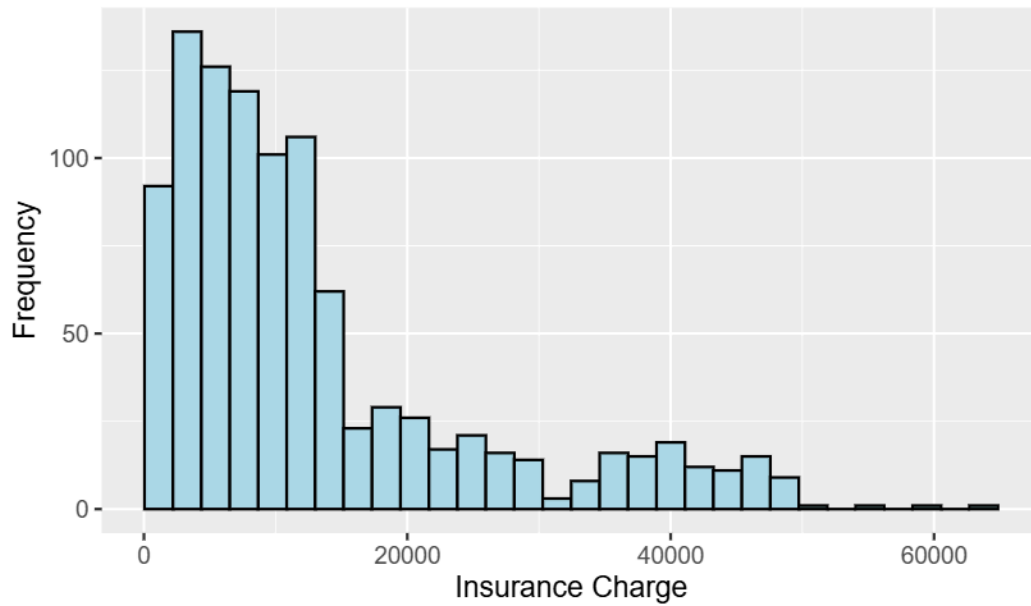
Understanding these patterns could help insurers, policymakers, and everyday people make better decisions about healthcare and insurance. From what I've read and observed in the past, it seems likely that insurance costs go up as people get older, and that men and women might have different cost patterns due to differences in health risks or medical use. My hypothesis is that age will be positively related to insurance costs, and that there will be noticeable differences between males and females.

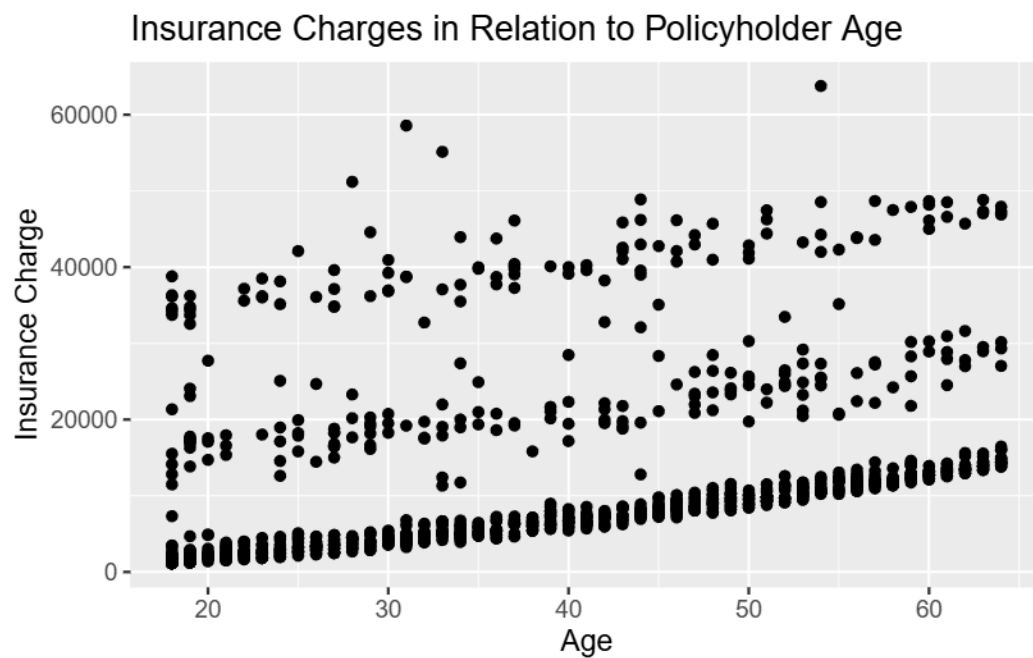
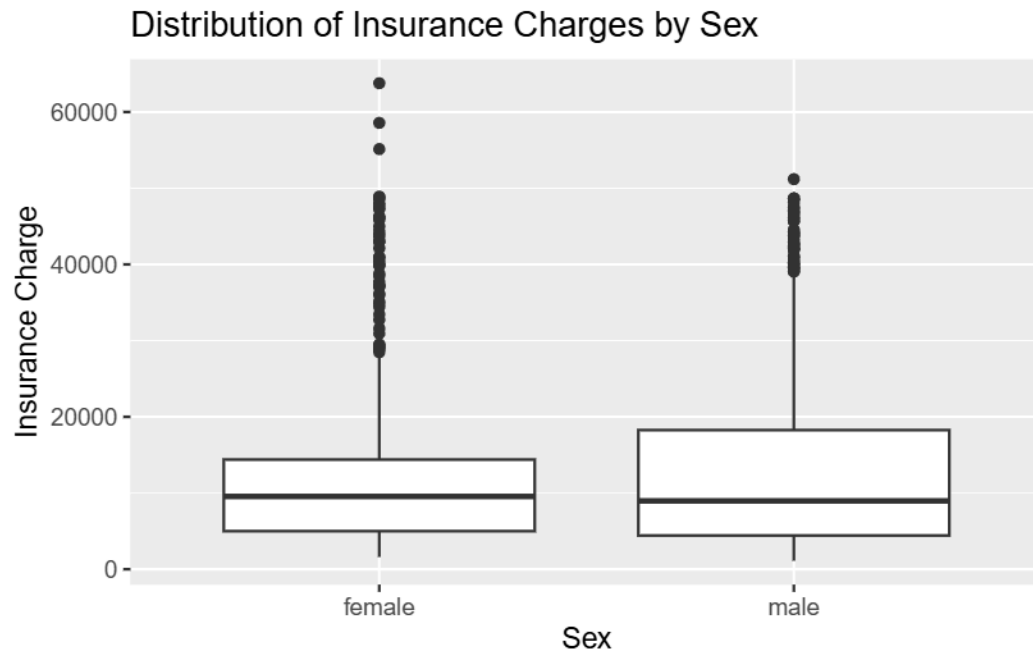
## Data

The dataset for this project consisted of many individual medical insurance records that included information on age, sex, annual insurance charges, and other factors. Data exploration provided key information as to which numerical variables would be the most compelling to work with. A correlation plot showed a fairly significant correlation between insurance charges and age, so, this was explored further. Sex was also a particular aspect that I wished to look into, so that variable was investigated as well. Plots observing the general frequencies of insurance charges, charges based on sex, and charges based on age were all examined. From these graphs, these conclusions were drawn: the most frequent insurance charges were those less than ~ \$12,000, the IQR for insurance charges for men was larger than that of women - spanning higher charges overall, and insurance charges increased linearly with age.



Distribution of Insurance Charges





## Models

Age was treated as a continuous numerical predictor, sex was coded as a binary categorical variable (0 = female (baseline), 1 = male), and charges served as the response variable. To assess how age and sex relate to insurance costs, three regression models were created: (1) a

simple model predicting insurance charges from age and sex, (2) an interaction model that added an age  $\times$  sex term to test whether the effect of age differed by sex, and (3) a model predicting log-transformed charges from age and sex to address possible skewness. Model assumptions were evaluated using residual plots and model performance was compared using  $R^2$ , adjusted  $R^2$ , and other summary statistics, with significance determined using p-value.

For Model 1, age and sex were used as predictor variables because they are fundamental demographic factors that are likely to influence medical insurance costs. Age is expected to have a positive effect since healthcare expenses generally increase as people get older. Sex was included as a categorical variable because males and females may experience different healthcare needs, which could affect insurance charges. I decided not to include other variables such as BMI, smoking status, or number of children because the focus of this model is to examine the basic relationship between cost and demographic factors. I also did not include interaction or exponentiated terms in this first model in order to create a simple baseline model.

The residual plots suggest that linearity is reasonable but there may be some curvature with age. The histogram and Q-Q plot indicate strong skew in residuals, likely due to some very high-cost individuals. Overall, assumptions are moderately satisfied, but the model may not fully capture non-linearity in age or interactions with sex.

For Model 2, age, sex, and the interaction between age and sex were included. The interaction term was added in order to examine the effect of age on insurance costs that could differ between males and females; for example, costs may rise faster with age for one sex. This allows the model to capture differences in the slope of age for males versus females.

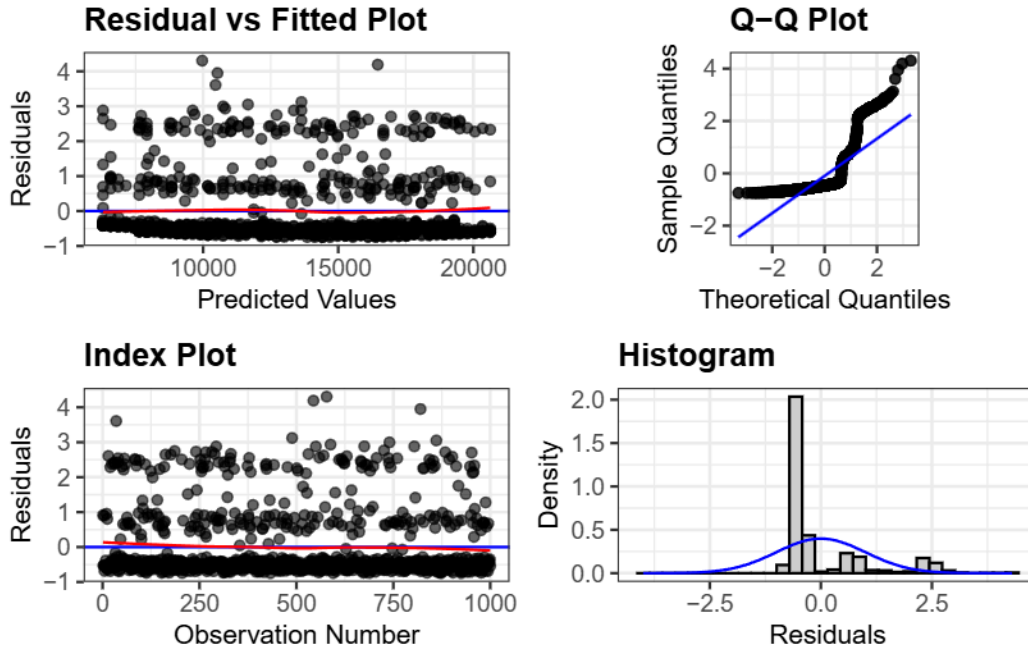
This model better captures differences in the effect of age by sex. Residual plots indicate mildly improved linearity for age within each sex, though skew still strongly exists. The Q-Q plot shows moderate deviation from normality. Overall, assumptions may be slightly better satisfied than Model 1, but the heavily skewed residuals suggest that transformation may be the best solution and next step to better fit the assumptions.

Model 3 used the same predictors as in Model 1 ('age' and 'sex') but applied a log transformation to the response variable 'charges'. The reason for this change is that the distribution of insurance costs is highly right-skewed. Taking the natural logarithm of charges helps stabilize the variance and makes the relationship between predictors and the response more linear. Quadratic or interaction terms were not implemented here in order to focus on evaluating whether the log transformation alone improves the model's assumptions and fit and also as to not over complicate the interpretation.

The log-transformed model noticeably improves the model assumptions. The residuals show a more random scatter, suggesting that the linearity assumption is better met than in the untransformed models. The histogram and Q-Q plot indicate that residuals are much closer to a normal distribution. This suggests that the log transformation was successful in stabilizing variance and improving the overall model fit.

Model 1:

$$\widehat{\text{Insurance Charge}} = b_0 + b_1 \times \text{Age} + b_2 \times \text{Sex}, \text{ where } \epsilon_i \sim \mathcal{N}(0, \sigma)$$



Call:

```
lm(formula = charges ~ age + sex, data = insurance)
```

Residuals:

| Min   | 1Q    | Median | 3Q   | Max   |
|-------|-------|--------|------|-------|
| -8446 | -6510 | -5275  | 4319 | 48605 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 1214.3   | 1139.4     | 1.066   | 0.2868     |
| age         | 282.3    | 25.3       | 11.161  | <2e-16 *** |
| sexmale     | 1340.5   | 715.8      | 1.873   | 0.0614 .   |

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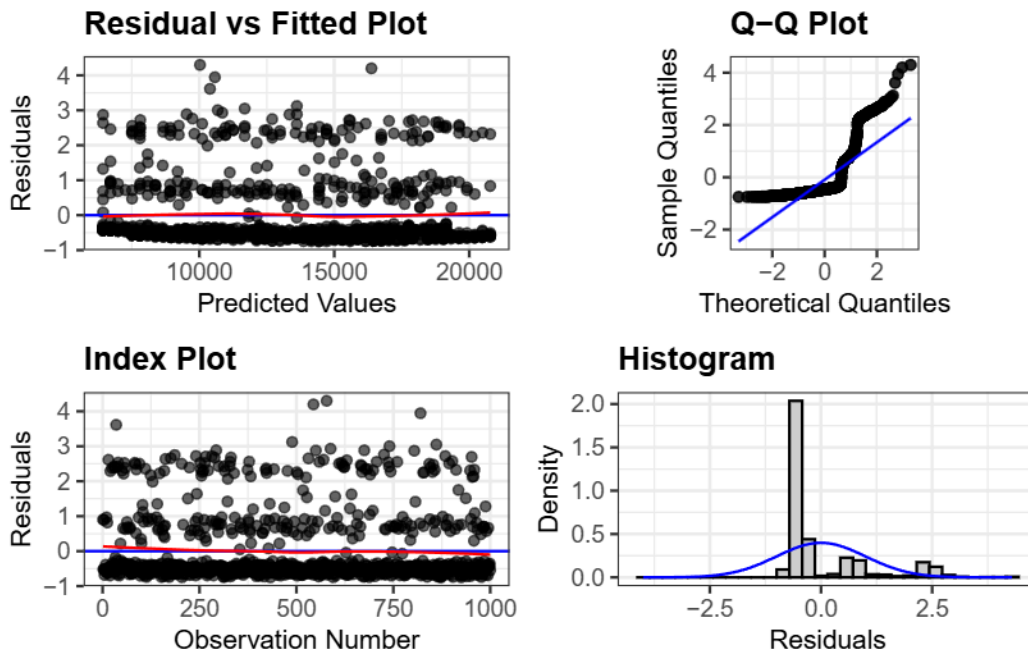
Residual standard error: 11300 on 997 degrees of freedom

Multiple R-squared: 0.1124, Adjusted R-squared: 0.1107

F-statistic: 63.16 on 2 and 997 DF, p-value: < 2.2e-16

Model 2:

$$\widehat{\text{Insurance Charge}} = b_0 + b_1 \times \text{Age} + b_2 \times \text{Sex} + b_3 \times \text{Age} \times \text{Sex}, \text{ where } \epsilon_i \sim \mathcal{N}(0, \sigma)$$



Call:

```
lm(formula = charges ~ age * sex, data = insurance)
```

Residuals:

| Min   | 1Q    | Median | 3Q   | Max   |
|-------|-------|--------|------|-------|
| -8469 | -6511 | -5261  | 4391 | 48548 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )     |
|-------------|----------|------------|---------|--------------|
| (Intercept) | 1459.03  | 1527.33    | 0.955   | 0.340        |
| age         | 276.26   | 35.72      | 7.733   | 2.56e-14 *** |
| sexmale     | 857.72   | 2129.48    | 0.403   | 0.687        |
| age:sexmale | 12.19    | 50.62      | 0.241   | 0.810        |

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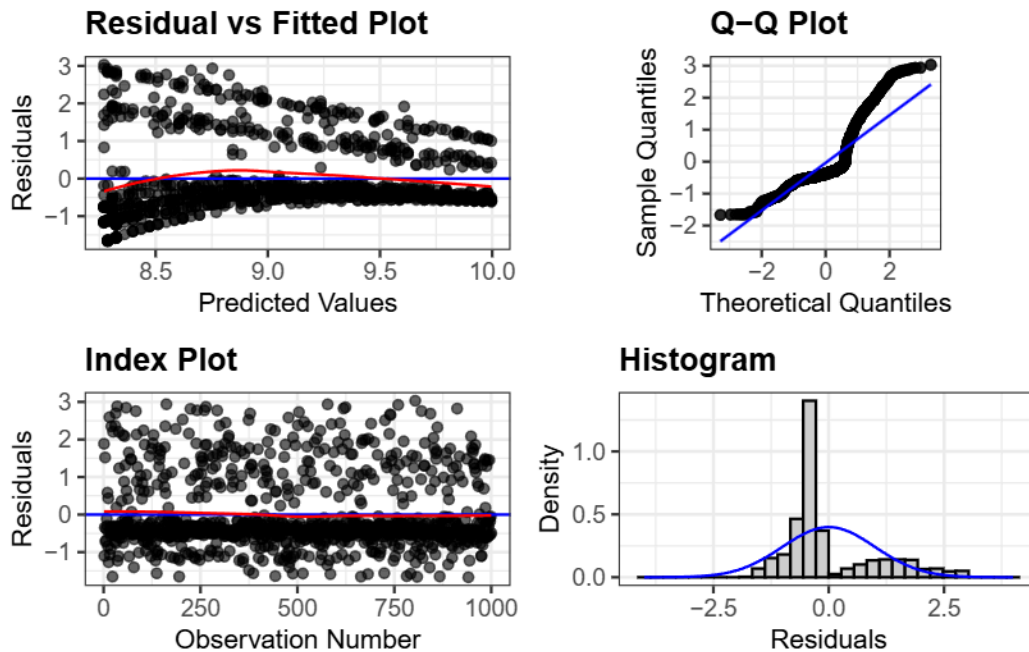
Residual standard error: 11310 on 996 degrees of freedom

Multiple R-squared: 0.1125, Adjusted R-squared: 0.1098

F-statistic: 42.09 on 3 and 996 DF, p-value: < 2.2e-16

Model 3:

$$\log(\text{Insurance Charge}) = b_0 + b_1 \times \text{Age} + b_2 \times \text{Sex}, \text{ where } \epsilon_i \sim \mathcal{N}(0, \sigma)$$



Call:

```
lm(formula = log(charges) ~ age + sex, data = insurance)
```

Residuals:

| Min     | 1Q      | Median  | 3Q     | Max    |
|---------|---------|---------|--------|--------|
| -1.2640 | -0.4097 | -0.2968 | 0.3518 | 2.2960 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 7.601268 | 0.076540   | 99.311  | <2e-16 *** |
| age         | 0.037151 | 0.001699   | 21.862  | <2e-16 *** |
| sexmale     | 0.016772 | 0.048082   | 0.349   | 0.727      |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7593 on 997 degrees of freedom

Multiple R-squared: 0.3243, Adjusted R-squared: 0.3229  
F-statistic: 239.2 on 2 and 997 DF, p-value: < 2.2e-16

## Results

Model 1 explains a small portion of the variation in medical insurance charges, with an  $R^2$  of 0.1124 and adjusted  $R^2$  of 0.1107 meaning age and sex together account for about 11% of the variability in insurance charges. This suggests that the variation is driven by other unmodeled variables. The F-statistic is significant ( $p < 2.2e-16$ ), indicating that this model predicts charges more than a null model. Age is a strong and highly significant predictor ( $\beta = 282.3$ ,  $p < 2e-16$ ). For each additional year of age, insurance charges increase by about \$282 on average, holding sex constant. Sex (male) shows a marginal effect ( $\beta = 1340.5$ ,  $p = 0.0614$ ), meaning males are estimated to have about 1,340 dollars higher charges than females on average, this p-value is close to the 0.05 threshold, suggesting weak evidence of sex differences in charges. The residual standard error (11,300) is large, confirming the model leaves substantial unexplained variation. The residual summary (ranging from -8446 to 48605) also suggests that charges have a long right tail, supporting the idea of trying a log-transformed model (Model 3), to improve fit.

Model 2 introduces an interaction between age and sex to allow the relationship between age and charges to differ between males and females. However, the model fit is virtually unchanged from Model 1:  $R^2 = 0.1125$  (versus 0.1124 in Model 1). Adjusted  $R^2 = 0.1098$  (slightly lower than Model 1's 0.1107 due to the extra parameter). Residual standard error = 11310, which was almost identical that of Model 1 (11300). This shows that including the interaction barely affects the model's explanatory power. The age  $\times$  sex interaction is not significant ( $\beta = 12.19$ ,  $p = 0.810$ ). This indicates that the slope of age on charges does not differ meaningfully between males and females. With all of this considered, Model 2 offers no definite advantage over Model 1. The effect of age on charges is similar for males and females, so no interaction is needed.

Model 3 improves dramatically over the earlier models.  $R^2 = 0.3243$ , compared with ~0.11 in Models 1 and 2. The adjusted  $R^2 = 0.3229$  while the F-statistic = 239.2, with a p-value < 2.2e-16. Explaining 32% of variance in log-charges versus only 11% in the previous models indicates that the log transformation substantially improves model fit, reduces skew, and produces a more linear, stable relationship with age. The high F-statistic with an extremely low p-value suggests that this model makes much improvement in prediction compared to the null model. The residuals also show a major improvement with a smaller range (-1.26 to 2.30), much more symmetric, and a lower residual standard error (0.7593 vs. 11,300 in raw-scale model). This confirms that the log scale better satisfies linear model assumptions. The age coefficient estimate = 0.03715 with a p-value < 2e-16. This means that a one-year increase in age is associated with a 3.7% increase in charges, on average (since  $\exp(0.037) \approx 1.038$ ). This means costs rise at a relative rate that becomes more meaningful as age increases. Males have ~1.7% higher charges than females on average, but this difference is not meaningful statistically or practically.

## Conclusion

This analysis showed that age is a strong and consistent predictor of medical insurance charges, while sex has little meaningful impact, and the relationship between age and costs does not differ between males and females. The log-transformed model fit the data best, indicating that insurance charges increase at a roughly proportional rate as people age and that medical cost data are better modeled on a log scale.

These findings suggest that future predictions of health insurance costs should place primary emphasis on age and use modeling approaches that account for skewed cost distributions. Since sex adds little predictive value, more detailed variables would likely contribute more to accurate forecasting and model building. Overall, using log-based models and focusing on age-related risk can improve the reliability of insurance cost predictions.

## Citations/References

Dataset: Medical Cost Personal Datasets - <https://www.kaggle.com/datasets/mirichoi0218/insurance>